

Builder Persona

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Living document - updated each chat session | Last updated: wfg-plans Session 5 - 26 Mar 2026 SGT

How to use this document: This is a living capability profile. Each chat session adds a dated layer. Timestamps in section headers record when each entry was made. At the start of each new chat, upload this document alongside the Builder Framework, Builder Lens, and handover document so Claude can read the full context.

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1. Synopsis

Chat 4 - 17 Mar 2026 SGT | Updated Chat 6 - 21 Mar 2026 SGT | Updated v2 build day - 23 Mar 2026 SGT | Updated wfg-plans Session 4 - 26 Mar 2026 SGT | Updated wfg-plans Session 5 - 26 Mar 2026 SGT

Field	Value
Name	Adrian K. L. Ang
Location	Singapore
Project	takearoundabout.com - AI Skills Analyser + wfg-plans NTUC-ARU Workforce Planning Dashboard
Current version	wfg-plans v0.1.5-beta - 26 Mar 2026 SGT
Profile type	Practitioner-builder - domain expert using AI as development partner
Starting point	Prior IT background, rusty systems knowledge, zero AI building experience at project start
Sessions completed	13+ sessions including independent code audit - approximately 100+ total active hours

Current state	Building and deploying wfg-plans v0 - 9-tab workforce planning dashboard, Vercel deployment, GitHub repository setup
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Adrian K. L. Ang is a Singapore-based workforce transformation practitioner who has now built two production AI-native applications from zero AI building experience at the start. The second project, wfg-plans, demonstrates a significant step up: the data source is a live organisational EAC register, the analysis is gold-standard workforce planning methodology, and the UI architecture is as complex as a commercial HR analytics product.

What makes this profile unusual is the combination: prior IT and coding experience (not a beginner), but genuinely no AI building knowledge at the start. The technical foundations were dormant, not absent. The project reactivated them and added a new layer - prompt engineering, agentic workflow design, AI product thinking - that did not exist before.

Chat 4 added meta-awareness: the builder can now name what he does, not just do it. Chat 6 added structured process discipline, documentation consolidation, and user-testing rigour through real observations from a Deputy CEO's session. v2 build day added REST API integration, parallel async data pipelines, and real-time incident response. wfg-plans Session 5 added operational deployment discipline: Git repository initialisation, GitHub Desktop publish, iCloud path resolution, and stable terminal command conventions.

2. Identity and Context

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Professional identity

Practitioner at the intersection of workforce transformation, AI readiness, and organisational development. Not a developer by primary identity - a workforce practitioner who builds because the question demanded it.

Why he builds

Not for credentials. Not for revenue. The underlying question - which skills AI is reshaping, and which remain distinctly human - deserved an honest, accessible answer. The knowledge should not sit behind a paywall or inside a consultancy deck.

Singapore context

Acutely aware of the anxiety landscape in Singapore around AI, economic security, and personal relevance. Understands the culture of short attention spans, negative framing of change, and the pressure to remain relevant in a fast-moving labour market.

Core belief about AI

Machines will not overtake humans because humans created them and continue to evolve alongside them. AI is a collaborator, not a replacement. This is not wishful thinking - it is a position held with evidence and examined through the lens of two years of cognitive bias research.

Dyslexia and identity

Publicly disclosed dyslexia for the first time in a professional context - included in LinkedIn profile About section as a formative influence on thinking, not as a vulnerability. - Chat 2 - 17 Mar 2026 SGT

Faith and values

Follower of Christ, self-described as flawed. Values of grace and mercy identified as underlying drivers of product and communication decisions. Kept implicit in public-facing content rather than named explicitly.

3. Professional Background

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Domain expertise

- HR and people development
- Workforce transformation and organisational development
- Learning and development
- Strategy and organisational consulting
- IT training and instruction
- IT consulting and infrastructure
- IT software and startup experience
- Governance experience

Governance experience

11 years as board member of Precept Ministries Singapore - governance, portable medical benefits, reserves policy, remuneration benchmark assessment. Never mentioned as a credential. The decision not to use board experience as a professional signal is itself a values decision.

Relevant practitioner knowledge

- Understanding of how AI is affecting the labour market at the occupational skills level

- ESCO v1.2 occupational taxonomy - not just as a data source but as a framework for thinking about skills
- Singapore workforce context - SkillsFuture, WSQ, MOM, NTUC, SSG ecosystem
- IMDA Model AI Governance Framework and EU AI Act awareness
- Hyper Island background - master's thesis on cognitive bias, structured analytical techniques, decision-making research

Working style

- Cross-disciplinary by nature - moves between HR, technology, strategy, and design without compartmentalising
- Builds from genuine questions, not from briefs
- Comfortable with ambiguity - makes decisions under uncertainty without paralysis
- Values precision in language - the copy decisions in this tool reflect professional-grade communication craft

4. Technical Background

17 Mar 2026 SGT | Extended v2 build day - 23 Mar 2026 SGT | wfg-plans Session 4 - 26 Mar 2026 SGT | wfg-plans Session 5 - 26 Mar 2026 SGT

Prior to this project

- Prior coding experience - not a beginner, but knowledge was rusty
- IT infrastructure background - understands networks, servers, hosting, hardware constraints
- IT software experience - understands software development lifecycle, not just end-user perspective
- IT startup experience - understands the full stack of building and shipping a product
- Systems mental model existed but needed reactivation - frontend/backend/API/database/hosting relationships understood in principle
- Zero AI building experience - the AI layer was genuinely new territory

Technical knowledge reactivated in this project

- React 18 component architecture - hooks, state management, useCallback, useRef, useEffect
- Vite build pipeline - esbuild, JSX compilation, production bundling
- Vercel serverless functions - api/claude.js as a proxy, maxDuration, environment variables
- HTTP security headers - CSP, X-Frame-Options, Permissions-Policy
- CSS responsive design - breakpoints, inline styles vs CSS classes, Retina scaling
- JSON parsing and error recovery - extractJSON pattern, token truncation handling

Technology limitations understood - original project

- Claude has no memory between sessions - continuity requires explicit context injection
- Inline px font sizes in React cannot be overridden by CSS breakpoints
- iOS Safari auto-zooms inputs below 16px font size - no CSS workaround
- Vercel free tier has concurrency limits - Fluid Compute resolves this
- Token budgets are hard limits - truncated JSON is a real failure mode, not a theoretical one
- Haiku 4.5 prompt caching requires minimum 4,096 tokens - below threshold causes API errors
- Claude generates from training knowledge, not live database - ESCO results are best-effort, not exhaustive
- Chat session compaction silently removes code defined in earlier sessions - browser is the only honest source of truth - Chat 3 - 17 Mar 2026 SGT
- React inline style objects do not support CSS functions - clamp(), calc(), min(), max() fail silently at runtime - Chat 3 - 17 Mar 2026 SGT
- Module-level JSX constants that reference undefined variables crash the app at load time - Chat 3 - 17 Mar 2026 SGT
- Vercel Analytics loads scripts from va.vercel-scripts.com - incorrect CSP domain silently blocks analytics - Chat 3 - 17 Mar 2026 SGT
- esbuild reports false line numbers when template literals span multiple lines - real cause is always an unclosed JSX div near a .map() closing brace - Chat 3 - 17 Mar 2026 SGT
- Two-call model split: rateSkills split into Haiku and Sonnet to stay within combined context limit - a single overloaded call fails silently with an infinite spinner - Chat 6 - 18 Mar 2026 SGT
- Token ceiling is silent: when system prompt plus output budget approaches a model combined token limit, the call fails with no useful error - Chat 6 - 18 Mar 2026 SGT
- Orphaned code after refactoring crashes the module at load time - Chat 6 - 18 Mar 2026 SGT
- Race conditions in async calls that share state: every async call writing shared state needs a cancellation flag - Chat 6 - 18 Mar 2026 SGT
- Claude artifact sandbox and Vercel serverless are two different execution environments with different API access rules - Chat 4 - 17 Mar 2026 SGT
- Vercel negative lookahead rewrite /(?!api).*) is essential - without it SPA rewrite intercepts serverless function routing and causes 404 on all API calls - Chat 4 - 17 Mar 2026 SGT
- Python string escaping in Claude computer-use scripts: \n written as literal newline inside JS strings causes build error - use binary replacement - Chat 6 - 19 Mar 2026 SGT
- React state props must be explicitly passed to child components - pickerFullError caused ReferenceError on live site - Chat 6 - 19 Mar 2026 SGT
- Single .catch() on async background calls is too aggressive - use retry loops with backoff instead - Chat 6 - 21 Mar 2026 SGT
- LLM models are unreliable for fixed lookup tasks with a bounded input space. When the input set is knowable, a client-side hardcoded lookup is more reliable - Chat 6 - 21 Mar 2026 SGT
- Concurrent async calls writing shared state require a cancellation ref pattern - Audit session - 22 Mar 2026 SGT
- Duplicate useEffect blocks with identical logic and empty dependency arrays register two event listeners on the same target - Audit session - 22 Mar 2026 SGT
- User input passed into Claude prompts via template literal is a prompt injection surface - apply a length cap, character validation, and a system prompt on every call - Audit session - 22 Mar 2026 SGT

- Child component props used without default values are a latent ReferenceError waiting for a second call site - Audit session - 22 Mar 2026 SGT

Technology limitations understood - v2 additions

- External REST APIs can return unexpected results for ambiguous queries - exact-match logic required over positional assumptions. ESCO search returned building caretaker as top result for Facilities Manager. - v2 Phase 1 - 23 Mar 2026 SGT
- Inline React px font sizes cannot be overridden by CSS media query breakpoints without !important on every property - global zoom or font raise in source is the only reliable approach. - v2 build day - 23 Mar 2026 SGT
- CSS zoom works on Chrome and Safari but not Firefox. For a px-based React app, raising font sizes in source is more reliable than CSS zoom. - v2 build day - 23 Mar 2026 SGT
- Heredoc paste (cat > file << EOF) can hang the terminal if the shell interprets special characters. Provide downloadable files instead. - v2 build day - 23 Mar 2026 SGT
- MacOS Safari auto-unzips downloaded zip files when Open safe files after downloading is enabled. - v2 build day - 23 Mar 2026 SGT
- API keys exposed in chat are compromised immediately and must be rotated - even if no apparent misuse. - v2 build day - 23 Mar 2026 SGT
- Global sed font-size substitutions cascade if run in sequence. Always verify distribution before and after. Restore from last known good zip if corrupted. - v2 build day - 23 Mar 2026 SGT

Technology limitations understood - wfg-plans additions

- esbuild 'Unterminated regular expression' error is a JSX structural parse failure, not a regex issue. Cause is unclosed map callback nesting. - wfg-plans Session 4 - 26 Mar 2026 SGT
- Non-ASCII characters (em dash, dagger, triangle symbols, arrows) in JSX string literals cause esbuild parse failures even inside quoted strings. Replace all with ASCII equivalents before packaging. - wfg-plans Session 4 - 26 Mar 2026 SGT
- Vercel MIME error (text/html is not a valid JavaScript MIME type) is a downstream symptom of a failed build. Diagnose npm run build first, not vercel.json routing config. - wfg-plans Session 4 - 26 Mar 2026 SGT
- React array returns from render functions: returning an array of tr elements from a function called inside tbody renders correctly without a Fragment wrapper. - wfg-plans Session 4 - 26 Mar 2026 SGT
- Absolute-positioned dropdown menus are clipped by overflow:hidden on ancestor containers. - wfg-plans Session 4 - 26 Mar 2026 SGT
- iCloud Drive folders accessed from MacBook terminal require the full path: ~/Library/Mobile Documents/com~apple~CloudDocs/... - wfg-plans Session 5 - 26 Mar 2026 SGT
- GitHub Desktop cannot see a folder as a repository unless .git exists inside that specific folder. git init must be run inside the project folder, not a parent. - wfg-plans Session 5 - 26 Mar 2026 SGT
- GitHub email privacy setting blocks terminal pushes. Use noreply address in git config or temporarily disable the setting to push. - wfg-plans Session 5 - 26 Mar 2026 SGT

5. AI Building - Knowledge and Skills

17 Mar 2026 SGT | Extended v2 build day - 23 Mar 2026 SGT | wfg-plans Session 4 - 26 Mar 2026 SGT

Prompt engineering

- Multi-field structured JSON prompts with compact token budgets
- System vs user prompt split for caching architecture
- Token budget calibration - understanding the relationship between budget and output quality
- Consistency rules in prompts - e.g. NA/HUMAN rule to prevent logical contradictions
- Prompt tone engineering - warm, humble, non-prescriptive language
- Structured output design - teaching Claude what to do when output is ambiguous or wrong
- Parallel prompt orchestration - 7+ calls per analysis, sequential and parallel
- Two-layer enforcement pattern - instruction in the prompt plus hard override in post-processing code. Prompts alone cannot be trusted for critical logic. - Chat 4 - 17 Mar 2026 SGT
- 19-technique prompt taxonomy: designed and applied a taxonomy of 19 prompting techniques with selection rules by skill type and automation level - Chat 6 - 18 Mar 2026 SGT
- Cross-batch technique diversity: cross-batch tracking via shared usedTechs Set prevents model defaulting to 3-4 safe techniques - Chat 6 - 18 Mar 2026 SGT
- Worked examples for underused techniques: abstract descriptions do not change model behaviour; concrete structural templates in the system prompt do - Chat 6 - 18 Mar 2026 SGT
- Plain language generation as a separate quality standard: prompts written for someone who has never used AI automation before - Chat 6 - 18 Mar 2026 SGT

Agentic workflow design

- Sequential and parallel API call orchestration
- Stale closure handling via useRef - real engineering pattern, not a workaround
- Queue architecture for deferred comparison analysis
- State management across a multi-step user journey
- Prompt caching system design - injectCaching proxy pattern
- Sequential vs parallel comparison execution - staggered parallel calls cause rate limit collisions in preview environments - Chat 3 - 17 Mar 2026 SGT
- Silent failure modes in async orchestration - always update state regardless of success, mark failed items explicitly - Chat 3 - 17 Mar 2026 SGT
- Background enrichment pattern: results screen shown immediately after structural analysis; Sonnet runs in background via Promise.allSettled with per-batch patchBatch callbacks - Chat 6 - 18 Mar 2026 SGT
- Parallel call overlap: rateSkills and progression/crossover/context fire simultaneously after getSkills - saves 5-8 seconds off critical path - Chat 6 - 18 Mar 2026 SGT
- Background description generation pattern - generateSkillDescriptions() fires after merge, targets only skills missing ESCO description - v2 Phase 2 - 23 Mar 2026 SGT
- Fail-silent per-item pattern in parallel fetching - fetchSkillDetail() returns empty object on any error. No single API timeout breaks the full result. - v2 Phase 1 - 23 Mar 2026 SGT

AI product thinking

- Identifying failure modes from user perspective, not code review
- Designing for trust - indicative framing, not authoritative claims
- Prompt-as-product - output quality is the product quality
- Alignment between tool values and copy tone
- Understanding where AI inconsistency requires client-side safety nets
- Visual output validation - caught NA tool / AI-Assisted level contradiction from a screenshot alone, without reviewing code - Chat 4 - 17 Mar 2026 SGT
- Meta-awareness of AI quality degradation - recognised own collaborative performance was at 70-75% precision after a long session due to context window saturation - Chat 4 - 17 Mar 2026 SGT
- Taxonomy credibility as product value - skills linked to canonical ESCO URIs with reuse level, alternative labels, narrower skills, and broader concepts. AI rates, taxonomy defines. - v2 build day - 23 Mar 2026 SGT
- Data architecture as UX - the Skill Categories redesign surfaced ESCO reuse level data as an interactive insight panel. - v2 Phase 3 - 23 Mar 2026 SGT
- Badge restraint - ESCO v1.2 badge added and removed in the same session. UUID pill already communicates ESCO provenance more precisely. - v2 Phase 3 - 23 Mar 2026 SGT

AI governance

- IMDA Model AI Governance Framework alignment
- EU AI Act minimal risk classification
- ESCO CC BY 4.0 attribution
- Honest indicative framing built into design, not added as disclaimer

6. Product Development Understanding

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Product thinking

- Single job principle - every feature serves the one sentence that describes what the tool does
- Minimum viable version discipline - ships something working before adding complexity
- Progressive disclosure - show the minimum, reveal detail on demand
- The user does not owe the product their attention - every word must earn its place
- Specificity as product value - generic output at scale is worthless
- Recognised over-engineering risk in real time - decided against adding academic bias and SAT methodology to the UI
- Content architecture thinking - identified repetition across three UI locations. Each surface must do a different job
- Scope expansion under pressure - correctly held the line on three separate requests that would have required Phase 5 architecture changes during Phase 2. - v2 build day - 23 Mar 2026 SGT

- Think before build applied to font sizing - recognised that a global CSS zoom approach was fragile before building it. - v2 build day - 23 Mar 2026 SGT

Distribution thinking

- Understood that product caveats belong inside the tool, not in the launch post. Resisted front-loading disclaimers. - Chat 2 - 17 Mar 2026 SGT
- Identified that an unfamiliar URL creates a trust barrier in Singapore click culture - solved the distribution problem in the comment copy without changing the product. - Chat 2 - 17 Mar 2026 SGT

User understanding

- Four distinct audiences with different anxieties and expectations
- Anxiety landscape awareness - designing for people who may be unsettled by the content
- Cultural awareness of short attention spans and negative framing in Singapore context
- Second vs third person - the difference between prescribing and inviting reflection
- The confidence gap - understanding that not starting is usually about confidence, not knowledge

Iteration discipline

- Test in production early - preview is not reality
- Say do not build yet - think before committing to code
- Screenshot before fix - visual evidence is faster than description
- One change at a time when debugging - isolate variables
- The deployment battle principle - blank screen after clean build means missing module-level constant before anything else - Chat 3 - 17 Mar 2026 SGT
- Grep before packaging - verify C, LEVELS, PERSONA_CONFIG, claudeCall, extractJSON, searchOccupations, getSkills, rateSkills in App.jsx before creating the deployment ZIP - Chat 3 - 17 Mar 2026 SGT

Documentation discipline

- Documents maintained: journal, feature summary, feature full, technical, builder persona, builder lens, builder framework, project register, handover, readiness assessment
- Handover document as continuity mechanism across sessions
- Builder Persona as living capability profile - updated each session
- Platinum Compact Framework adopted - structured response format reduces token cost without information loss - Chat 6 - 21 Mar 2026 SGT
- Builder Framework updated with Section 8 (Deliverable File Naming). Additive only - no prior content removed. - wfg-plans Session 5 - 26 Mar 2026 SGT
- Three documents updated in single session: Builder Framework, Journal, Timesheet. - wfg-plans Session 5 - 26 Mar 2026 SGT

7. System and Technology Understanding

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Architecture understanding

- Frontend / backend / API / serverless function relationship - clear mental model
- Client-side vs server-side execution - understands what runs where and why
- Environment variables - knows why API keys must never be in client code
- Build pipeline - source code to production bundle to deployed URL
- DNS and domain management - custom domain to Vercel project assignment
- Preview vs deployment runtime distinction - Claude artifact sandbox and Vercel serverless are two different execution environments with different API access rules - Chat 4 - 17 Mar 2026 SGT

Hardware and infrastructure awareness

- Server-side execution constraints - Vercel function timeout limits
- Network latency - understands why parallel calls are faster than sequential
- Mobile hardware constraints - iOS rendering differences, viewport handling, font scaling
- Retina display scaling - CSS pixel vs physical pixel, why 1280px CSS = 2560px physical
- Concurrency limits - API rate limits, Fluid Compute, Tier 2 activation

Software understanding

- Deployment lifecycle - source, build, bundle, deploy, rollback
- Version management - immutable deployments, separate projects for major versions
- Dependency management - npm, package.json, lock files
- Security layers - HTTP headers, CSP, serverless proxy pattern
- Error handling - warm error messages, graceful degradation, user-facing vs debug messages
- npm global package permissions on macOS - EACCES errors when updating globally installed CLI tools - Chat 3 - 17 Mar 2026 SGT
- Vercel plan features matter at deployment time - know plan constraints before designing deployment strategy - Chat 3 - 17 Mar 2026 SGT
- package-lock.json pins exact dependency versions for Vercel build environment - keep the lock file - Chat 3 - 17 Mar 2026 SGT

8. Human-Machine Interfacing

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UX philosophy

- The search box is the primary action - it must be the first thing the user sees

- Reduce to the minimum that works - cut every redundant instruction
- Colour encodes meaning, never decoration - semantic colour system enforced
- Mobile first, always - designed for 375px before 1440px
- Progressive disclosure - show the minimum, reveal detail on demand

Copy and tone as interface

- Third person neutral removes presumption - the tool does not tell people what to do
- Timing claims removed - never promise what the system cannot guarantee
- Indicative framing - starting point for reflection, not a verdict
- Warm error messages - the system speaks like a thoughtful colleague, not an error code
- Persona labels as invitations, not instructions - Fresh Graduates, not You as a fresh graduate

Data integrity as live communication decision

When writing the first comment post-launch, stopped to verify what data was actually collected before stating no data collected. Corrected the claim to only country of query, nothing else. Source of truth applies to one's own statements, not only to the tool's data sources.

AI collaboration as a skill set

Can name the skills applied across the project: prompt engineering, agentic workflow design, AI output validation, human-AI task division, iterative refinement, critical evaluation of AI suggestions, and the meta-skill of knowing when to start fresh.

Connected master's thesis on cognitive bias to the tool built - recognised the tool carries the same biases his research warned about. Intellectually honest about this without being paralysed by it. - Chat 4 - 17 Mar 2026 SGT

Accessibility and cultural sensitivity

- Singapore anxiety context - non-alarmist framing by design
- Short attention span culture - every screen must deliver value within the first scroll
- Multi-generational users - language accessible to graduates and senior practitioners alike
- Font sizing across device types - phone, tablet, laptop, 2K Retina all tested
- iOS-specific rendering bugs identified and resolved from device testing
- Semantic HTML, ARIA labels, skip navigation link, accessible search input - added after accessibility audit flagged Poor: html>body - Chat 6 - 18 Mar 2026 SGT

9. Cultural Awareness and Context

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Singapore workforce context

- Understands the anxiety around AI displacement - economic security is personal, not abstract
- Aware of the war and economic security framing that shapes how Singaporeans receive AI news
- Short attention span culture - content must earn its place in the first few seconds
- Negativity bias in the surrounding narrative - the tool deliberately counteracts this
- SkillsFuture, WSQ, MOM, NTUC ecosystem - understands the institutional landscape
- Singapore click caution - Singaporeans are wary of unfamiliar URLs and data collection. Crafted comment copy to address this: no sign up, no personal data collected, nothing linked back to you. - Chat 2 - 17 Mar 2026 SGT

ASEAN and global awareness

- ESCO taxonomy is EU-originated but applied to Singapore and ASEAN context
- O*NET is US-originated - US wage data caveat built into the tool design
- Singapore users need localised salary and career resources - MyCareersFuture, WSG
- Cultural directness - Singapore users appreciate clarity and brevity over elaborate framing

Organisational culture awareness

- Understands hierarchy and trust dynamics in Singapore workplaces
- Knows how HR and L&D practitioners receive tools - sceptical of vendor tools, open to peer-built ones
- Aware of the difference between individual and organisational AI readiness
- NTUC and union affiliate context - tool designed to be appropriate for labour movement use

10. Capability Ratings

Chat 4 - 17 Mar 2026 SGT | Updated Chat 6 - 21 Mar 2026 SGT | Updated v2 build day - 23 Mar 2026 SGT |
Updated wfg-plans Session 4 - 26 Mar 2026 SGT | Updated wfg-plans Session 5 - 26 Mar 2026 SGT

Scale: 0 = none | 1 = awareness only | 2 = basic | 3 = working | 4 = proficient | 5 = advanced

AI Building

Skill / Capability	Before Chat 1	After Chat 6	Notes
Prompt engineering - structured JSON	0/5	4/5	Core skill of this project
Prompt engineering - tone and framing	0/5	4/5	Values-driven from the start
19-technique prompt taxonomy	0/5	4/5	Designed and applied in Chat 6
Token budget calibration	0/5	3/5	Learned through production failures
Prompt caching architecture	0/5	3/5	Implemented and debugged in Chat 4
Agentic workflow design	0/5	3/5	7+ call orchestration built and shipped

AI product thinking	0/5	4/5	Domain expertise accelerates this
Visual output validation	0/5	4/5	Caught NA/AI-Assisted contradiction from screenshot
Meta-awareness of AI degradation	0/5	4/5	Recognised context saturation as structural
AI governance application	1/5	4/5	Existing awareness, now applied
Hardcoded lookup pattern	0/5	4/5	SENIOR_MGMT_LOOKUP: identified, designed, and applied in Chat 6
Evaluation and testing	0/5	1/5	Manual only - harness not yet built
RAG / retrieval patterns	0/5	1/5	Understood conceptually, not built
Code audit - reading and assessing AI-generated code	0/5	3/5	First full audit conducted - 14 findings, 17 fixes actioned - Audit session 22 Mar 2026 SGT

Frontend Engineering

Skill / Capability	Before Chat 1	After Chat 6	Notes
React hooks and state management	2/5	4/5	Rusty knowledge reactivated and extended
CSS responsive design	2/5	3/5	Inline vs class tension still present
Mobile-specific debugging	1/5	4/5	iOS bugs found and fixed from device testing
Performance and font scaling	1/5	3/5	2K/Retina addressed in Chat 4
Vite build pipeline	1/5	3/5	Working knowledge, not deep
Accessibility (ARIA, semantic HTML)	1/5	3/5	Raised from 2/5 after Chat 6 audit work

DevOps and Deployment

Skill / Capability	Before Chat 1	After Chat 6	Notes
Vercel project architecture	1/5	4/5	Multiple projects, domain management
Environment variable management	2/5	4/5	Clean from the start
Production debugging with logs	1/5	4/5	Vercel function logs used effectively
Vercel CLI	1/5	3/5	CLI now primary deployment method
Pre-deploy testing discipline	1/5	2/5	Still catching most issues post-deploy

Product and Design

Skill / Capability	Before Chat 1	After Chat 6	Notes
UX copy and tone	3/5	5/5	Domain expertise fully applied
User persona thinking	3/5	5/5	Strong from professional background
Feature prioritisation	3/5	4/5	Deferred items managed well
Distribution thinking	2/5	4/5	Applied under live conditions in Chat 2
Visual design and layout	2/5	3/5	Functional, not formally designed
Accessibility (a11y)	1/5	3/5	Semantic HTML, ARIA, skip nav added in Chat 6
User testing - qualitative	0/5	2/5	Deputy CEO session produced 6 targeted builds. Still solo and post-deploy.

AI Building - updates from v2 build day (23 Mar 2026)

Skill / Capability	After Chat 6	After v2	Notes
REST API integration and proxy design	0/5	4/5	Built, tested, debugged in Phase 1

Parallel async data pipeline	0/5	4/5	Promise.all across 36 skill URIs, fail-silent per item
Multi-layer data shape management	2/5	4/5	ESCO -> getEscoSkills -> merge -> state -> render - all fields carried explicitly
Taxonomy credibility design	1/5	4/5	ESCO URI links, reuse level, alt labels, narrower/broader surfaced as product value
Incident response - API key rotation	0/5	4/5	Rotated under pressure in under 5 minutes, no v1 downtime
Scope management under pressure	3/5	4/5	Held line on 3 Phase 5 items during Phase 2, documented and deferred correctly
External URL format discovery	0/5	3/5	4 formats tested, correct one confirmed by browser testing

AI Building - updates from wfg-plans Session 4 (26 Mar 2026)

Skill / Capability	After v2	After wfg-plans S4	Notes
REST API integration and proxy design	4/5	4/5	Unchanged - established in v2
Parallel async data pipeline	4/5	4/5	Unchanged - established in v2
Production debugging - JSX/esbuild	0/5	4/5	Root-caused three build failures in one session: JSX nesting, non-ASCII chars, Vercel MIME error
Pivot table architecture	0/5	4/5	Independent drill state, row-click navigation, chart-table sync, breadcrumb back-nav
Shared component prop design	2/5	4/5	getSubRows and onRowClick props - data logic stays in parent, display logic in component
Data grouping correctness	2/5	4/5	statsGroupedBySection - blank field values handled explicitly, no silent data loss
Hierarchical UI navigation	2/5	4/5	4-level scope drill, breadcrumb nav, expand/collapse within tables
Large dataset React architecture	2/5	4/5	1,521-row embedded dataset, all computations live from RAW_ESTAB, no stale data
Scope management under pressure	4/5	5/5	Held canonical hierarchy line, confirmed scope before every build, zero scope drift this session
Documentation discipline	4/5	5/5	All 13 documents updated, naming convention applied, handover complete

AI Building - updates from wfg-plans Session 5 (26 Mar 2026)

Skill / Capability	After S4	After S5	Notes
Git workflow and repository management	0/5	3/5	Initialised local repo, resolved remote conflicts, PAT authentication, GitHub Desktop publish, iCloud path handling. Push not yet fully complete.
Vercel deployment via CLI vs GitHub	3/5	4/5	Diagnosed MIME root cause as build failure not routing; understands CLI vs GitHub import paths; environment variable management.
Directed AI session management	3/5	4/5	HDR protocol audit, serialisation enforcement, document naming

			convention, zip filename discipline - all self-corrected within session.
Build trail documentation	4/5	5/5	Three documents updated in single session: Builder Framework (new section), Journal (addendum), Timesheet (new session row). Additive discipline maintained.

11. Builder's Beliefs and Philosophy

17 Mar 2026 SGT | Updated across Chats 2-6 | Updated v2 build day - 23 Mar 2026 SGT | Updated wfg-plans Session 4 and Session 5 - 26 Mar 2026 SGT

On AI | 17 Mar 2026 SGT

Machines will not overtake humans because humans created them and continue to evolve with them. AI is a collaborator. The practitioner provides judgment. The AI accelerates execution. The question of which skills remain distinctly human is worth asking honestly.

On building | 17 Mar 2026 SGT

Build because the question matters, not because the credential helps. Ship something working before adding complexity. The scenic route teaches more than the shortcut.

On users | 17 Mar 2026 SGT

The user does not owe the product their attention. They are capable of making their own decisions. Do not prescribe. Show the landscape. Respect the judgment.

On knowledge | 17 Mar 2026 SGT

Public knowledge should not sit behind a paywall. Free and useful is a meaningful act. Each query costs something - that is the honest reality of running this as a solo project.

On honesty | 17 Mar 2026 SGT

Indicative, not authoritative. Starting point, not verdict. This is a design principle, not a legal disclaimer.

On data integrity | Chat 2 - 17 Mar 2026 SGT

Source of truth applies to one's own statements, not only to the tool's data sources. When a minor inaccuracy in a public comment was identified, it was corrected immediately, not explained away.

On grace and mercy | Chat 2 - 17 Mar 2026 SGT

Built something free, asked nothing of the user, pointed the closing line of the second LinkedIn post toward someone else who might need it. These are not marketing decisions. They are values decisions.

On failure as signal | Chat 3 - 17 Mar 2026 SGT

A blank screen after a clean build is not a code error - it is a context error. Something the app expects is not there. The build tool cannot see it. The browser can.

On compaction | Chat 3 - 17 Mar 2026 SGT

Chat session memory is not infinite. Code defined in Chat 1 may not survive to Chat 4. The builder's responsibility is to carry what matters forward explicitly.

On restraint | Chat 4 - 17 Mar 2026 SGT

A tool's honesty is built into its design. If the design is already honest, explaining the honesty adds length without adding trust. Less is more when the work speaks.

On knowing when to stop | Chat 4 - 17 Mar 2026 SGT

Knowing when a conversation is too long to continue well is itself a skill. Starting fresh is not retreat - it is precision. A degraded session produces degraded work.

On academic knowledge in practice | Chat 4 - 17 Mar 2026 SGT

A practitioner who researched cognitive bias for two years built a tool that carries the same biases he studied. That is not hypocrisy - it is honesty. The tool is indicative by design because of what the research found, not in spite of it.

On process discipline | Chat 6 - 21 Mar 2026 SGT

A framework is only useful if it reduces cost without reducing quality. The Platinum Compact Framework does this: structured header, no prose, download links only. Every field earns its place.

On user testing | Chat 6 - 21 Mar 2026 SGT

A single real user session - a Deputy CEO trying the tool for the first time - produced six targeted builds in one evening. Feedback from observation is more precise than feedback from assumption.

On incident response | v2 build day - 23 Mar 2026 SGT

The API key was exposed in a chat message. The correct response was immediate, calm, and complete: identify, rotate, update both projects, verify v1 continuity. No drama. No delay. A security incident handled well is also a product decision - it protects the users who depend on the tool.

On taxonomy as truth | v2 build day - 23 Mar 2026 SGT

AI generates. Taxonomy defines. The most credible thing the tool can say about a skill is not that Claude thought it was relevant - it is that the ESCO v1.2 REST API returned it as essential for this occupation. The AI's role is to rate and interpret. The taxonomy's role is to establish ground truth.

On font sizes and decisions | v2 build day - 23 Mar 2026 SGT

Raising font sizes globally to solve a readability problem on large screens breaks the layout on the device the tool was designed for first. A fix that helps one user at the cost of another is not a fix - it is a trade-off.

On data accuracy as product integrity | wfg-plans Session 4 - 26 Mar 2026 SGT

A workforce planning dashboard that silently hides a Director post because it has no section assigned is not a planning tool - it is a misleading one. Every estab record must be visible at every drill level, regardless of how the source data was classified. The tool serves the planner, not the data structure.

On canonical hierarchy as non-negotiable UX constraint | wfg-plans Session 4 - 26 Mar 2026 SGT

The hierarchy is not a preference. Group > Division > Department > Section > Estab Title is the organisational reality. Every tab, every table, every label must reflect it. When a build drifts from this, it is not a UX issue - it is a data integrity issue.

On operational discipline as product decision | wfg-plans Session 5 - 26 Mar 2026 SGT

Keeping the deploy zip filename as WFG-Plans beta.zip across every session is not a minor convenience. It is a decision that the Lead should not carry cognitive load about filenames when deploying. Stable terminal commands are part of the product. Operational discipline is product thinking applied to the build process itself.

12. Development Preferences

17 Mar 2026 SGT | Updated Chat 6 - 21 Mar 2026 SGT

Stack

- JavaScript / React. Single JSX file. No separate CSS unless scale demands it.
- Vercel hosting. CLI deployment (vercel --prod).
- Claude Haiku 4.5 for production. Anthropic API via serverless proxy.
- Claude Sonnet 4.6 for sophisticated prompt generation - generatePrompts function only; all other calls remain on Haiku - Chat 6 - 18 Mar 2026 SGT
- Open, attributed public datasets - ESCO, O*NET, SkillsFuture.

Process

- Say do not build yet before every significant feature
- Deploy after each major feature - test in production early
- Share screenshots when something looks wrong
- Keep handover documents - factual, specific, written for the next Claude session
- Upload source files at the start of every new chat

- Browser console first - when a blank screen appears, open the browser console before touching any code - Chat 3 - 17 Mar 2026 SGT
- Grep check before packaging - search for C, LEVELS, PERSONA_CONFIG, claudeCall, extractJSON, searchOccupations, getSkills, rateSkills in App.jsx before creating the deployment ZIP - Chat 3 - 17 Mar 2026 SGT
- Start a new conversation when context window is saturated - do not continue a degraded session for work requiring careful judgment - Chat 4 - 17 Mar 2026 SGT
- Platinum Compact Framework as response standard - HDR block, one-line fields, download links only, no inline code - Chat 6 - 21 Mar 2026 SGT
- Serialisation line at the top of every response: No NNN - DD-MM-YY HH:MM | SN-NNN. HDR resets per chat; serial number carries forward across all sessions. - wfg-plans Session 5 - 26 Mar 2026 SGT
- Deploy zip filename fixed as WFG-Plans beta.zip - never serialised with version numbers or dates. - wfg-plans Session 5 - 26 Mar 2026 SGT

Design

- Clean, restrained. White surfaces, subtle borders, generous whitespace.
- Colour encodes meaning, never decoration.
- Mobile first, always. 375px before 1440px.
- Progressive disclosure. Expandable rows over separate pages.
- Consistent font sizes across equivalent content types.

13. Growth Path

Chat 4 - 17 Mar 2026 SGT | Updated Chat 6 - 21 Mar 2026 SGT | Updated wfg-plans Session 5 - 26 Mar 2026 SGT

Next capabilities to build

- Complete Vercel deployment via GitHub - wfg-plans-v0 repo live on Vercel with env vars configured
- Phase 4: credibility copy update throughout the UI to reflect v2 ESCO API direct source
- Phase 5: skill detail link + replace Who Uses This Skill with ESCO isEssentialForOccupation data
- Scope comparison panel for Etab by Grade/Title tab
- Verify scope comparison rows correct for all 7 metric cards (Workforce Statistics)
- R&R tab: confirm ESCO_LOOKUP coverage for top NTUC-ARU titles
- R&R tab: upgrade from hardcoded ESCO to live ESCO v1.2 API
- User research sessions - structured, not solo and post-deploy
- Problem brief discipline - repeatable practice, not intuitive only
- Evaluation criteria before shipping - define success before building
- Evaluation and testing harness - automated test cases for prompt quality
- Retrieval Augmented Generation - SkillsFuture, MOM, IBF as grounding sources
- Pre-deploy testing discipline - catch issues before live, not after
- PDF export - full role analysis and comparison report as downloadable documents

- Font size strategy for large screens - responsive font scaling without breaking mobile layout

Next articles worth writing

- The stale closure - silent failure and AI tools that look correct but are not
- The queue architecture - user agency and designing for what the user knows
- The honest builder - what it means to build a tool that carries the biases you spent two years researching
- The blank screen - what chat compaction taught about runtime vs compilation - Chat 3 - 17 Mar 2026 SGT
- Start Here - the gap between knowing and doing, and what it takes to close it

Platform direction

- Register sole proprietorship with ACRA when SSG Developer Portal access is needed
- NTUC partnership conversation when tool gains meaningful traction
- O*NET integration for occupation and skills data depth
- Singapore-localised salary and career resources alongside US-origin data

For the next chat session

Upload this document alongside the Builder Framework, Builder Lens, and handover document. Ask Claude to read the Builder Persona before any work begins. Add a new stamped entry to this document at the end of each session.

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Sometimes the scenic route is the right one.

Session 6 Additions - Added 28-3-26 1201 SGT

Synopsis Update

Added 28-3-26 1201 SGT

Field	Value
Projects	takearoundabout.com - AI Skills Analyser wfg-plans - OUVD-ARU Workforce Planning Dashboard (redacted from NTUC-ARU - 28-3-26 SGT)
Current version	wfg-plans v0.2.1-beta - 28 Mar 2026 SGT

Current state	v0.2.1-beta deployed live to Vercel. Redaction complete. 9 tabs functional.
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Technology Limitations - Session 6 additions

Added 28-3-26 1201 SGT

- index.html <title> is separate from App.jsx. Redaction passes on App.jsx do not cover the window title. Must be updated independently. - wfg-plans S6
- sed replacements with plural/singular pairs: execute plural first. UNIONS before UNION prevents double-hit producing FORCESS. - wfg-plans S6
- git push via terminal (git add . && git commit && git push) is more reliable than GitHub Desktop when files are replaced by drag-and-drop. GitHub Desktop requires files to be inside the tracked folder before detecting changes. - wfg-plans S6
- README.md is a tracked file. Version numbers and titles in README.md must be updated at every version bump alongside App.jsx and index.html. - wfg-plans S6
- Document update method: unpack -> append XML -> repack. Generating documents from scratch loses all original styles, fonts, spacing, and paragraph formatting. Always use the prior session's .docx as the base. - wfg-plans S6

Capability Ratings - Session 6 additions

Added 28-3-26 1201 SGT

Skill / Capability	After S5	After S6	Notes
Large-scale data redaction	0/5	4/5	Six terms, 4,940 lines, zero residuals, side effects anticipated and confirmed
Version identity management	3/5	4/5	Beta 2 designation for identity-level change vs patch
Terminal git workflow	2/5	4/5	git add . && commit && push resolved deployment confusion that GitHub Desktop could not
Multi-file version consistency	2/5	4/5	App.jsx + index.html + README.md all updated together
Document update discipline	3/5	5/5	Unpack-append-repack confirmed as correct method. Source file as base always.

Builder's Beliefs - Session 6 additions

Added 28-3-26 1201 SGT

On redaction as product stewardship | wfg-plans Session 6 - 28 Mar 2026 - The dashboard was built with real data. Redacting it is not cosmetic - it changes the tool's public identity. That deserves a version number, not a patch tag. The right frame is: who does this tool say it belongs to, and who can it safely be shown to? Getting the answer right before deployment is product stewardship, not administrative overhead.

On operational clarity | wfg-plans Session 6 - 28 Mar 2026 - When file management becomes confusing between editor, GitHub Desktop, and terminal, the terminal wins. `git add . && commit && push` is the ground truth. The GUI is a convenience, not a guarantee.

On document integrity | wfg-plans Session 6 - 28 Mar 2026 - Generating a document from scratch looks complete but is smaller and loses prior content. Appending to the source file preserves everything. The right method is `unpack-append-repack`, not regenerate.